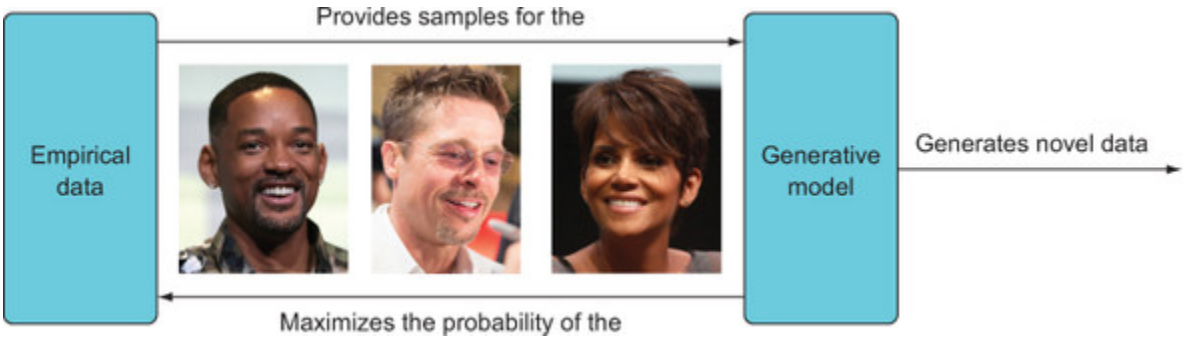


The images in data set *Q* were sampled from the real world, but they are just a small sample of the infinite number of photographs that already exist but are not in the data set and that could have been taken but were not. For example, there may just be one headshot photo of Will Smith in the data set, but another photo of Will Smith taken at a different angle could just as easily have been part of the data set. A photo of Will Smith with a baby elephant on top of his head, while not impossible, would be less likely to be included in the data set because it is less likely to exist (who would put a baby elephant on their head?).

There are naturally more and less likely photos of celebrities, so the real world has a probability distribution over images of celebrities. We can denote this true probability distribution of celebrity photos as $Q(x)$, where x is some arbitrary image, and $Q(x)$ tells us the probability of that image existing in the world. If x is a specific image in data set Q, then $Q(x) = 1.0$, since that image definitely exists in the real world. However, if we plug in an image that's not in the data set but likely exists in the real world outside of our small sample, then $Q(x)$ might equal 0.9.

When we randomly initialize our generative model P, it will output random-looking images that look like white noise. We can think of our generative model as a random variable, and every random variable has an associated probability distribution that we denote $P(x)$, so we can also ask our generative model what the probability of a specific image is given its current set of parameters. When we first initialize it, it will think all images are more or less equally probable, and all will be assigned a fairly low probability. So if we ask P("Will Smith photo") it will return some tiny probability, but if we ask Q("Will Smith photo"), we'll get 1.0.